Because a new term has been added to **H**, there will be a new term in the commutator of **X** and **H**. Potentially, that could modify the expression for velocity in Eq. 9.9, but it's easy to see that this doesn't happen. The new term involves the commutator of **X** with **V** (*x*). But multiplying by *x* and multiplying by a function of *x* are operations that commute. In other words,

$$[\mathbf{X}, \mathbf{V}(x)] = 0.$$

Therefore, the connection between velocity and momentum is unaffected by forces in quantum mechanics, as is the case in classical mechanics.

The more interesting question is: Can we understand the quantum version of Newton's law? As stated above, this law can be written as

$$\frac{dp}{dt} = F.$$

Let's calculate the time derivative of the expectation value of **P**. Again, the trick is to commute **P** with the Hamiltonian:

$$\frac{d}{dt}\langle \mathbf{P} \rangle = \frac{i}{2m\hbar}\langle [\mathbf{P}^2, \mathbf{P}] \rangle + \frac{i}{\hbar}\langle [\mathbf{V}, \mathbf{P}] \rangle. \qquad (9.15)$$

The first term is zero because an operator commutes with any function of itself. To compute the second term, we'll use an equation that we haven't proved yet:

$$[\mathbf{V}(x), \mathbf{P}] = i\hbar \frac{dV(x)}{dx}. \qquad (9.16)$$

Plugging Eq. 9.16 into Eq. 9.15, we get